



Vaibhav Vemani

Date of birth: 09/12/2003

Place of birth: Madanapalle, India

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Gender: Male

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WORK EXPERIENCE

BeyondRisX Bangalore, India

Software Engineering Intern

05/02/2026 - Current

- Architected and deployed a full-stack risk assessment platform for BeyondRisX, managing end-to-end system design and implementation.
- Developed a secure client portal with email OTP authentication and session management, allowing users to pause and resume assessments seamlessly.
- Engineered an admin dashboard for dynamic questionnaire creation and built a shareable assessment link generator to streamline test distribution.

GeolQ Bangalore, India

Software Developer Intern - Backend

22/07/2025 - 21/01/2026

- Reduced API latency by 93% (30 s $\rightarrow$ 2 s) by optimising complex PostgreSQL queries with strategic indexing and pagination for heavy spatial data layers.
- Built resilient payment infrastructure (Razorpay) with idempotent webhook processing and automated subscription credit expiration logic.
- Resolved GeoServer throughput bottlenecks in SpatialIQ (core map-rendering microservice) by introducing tile-level caching, geometry simplification at ingestion, and request queuing.

RoboDynamics Bangalore, India

Robotics Developer Intern

08/03/2023 - 08/08/2023

- Designed and deployed interactive IoT demonstration models, successfully driving a 30% increase in course enrollment and sales.
- Engineered a scalable educational software platform to optimize the learning experience, accelerating customer registrations by 25%.
- Integrated Machine Learning algorithms with IoT hardware to simulate real-world use cases, significantly boosting student engagement and hands-on interaction.

EDUCATION & TRAINING

13/10/2022 - CURRENT Bangalore, India

Bachelor of Engineering New Horizon College of Engineering

Website: <https://newhorizoncollegeofengineering.in> **Field(s) of study:** Artificial Intelligence and Machine Learning | **Final grade:** 9.12/10.0

10/06/2020 - 08/04/2022 Bangalore, India

Class XII Shri Ram Global School

Website: <https://srgsbangalore.com> | **Final grade:** 77%

10/08/2019 - 10/04/2020 Lewis Center, United States

Class X Olentangy High School

Website: <https://ohs.olentangy.k12.oh.us> | **Final grade:** 3.6/4

PROJECTS

01/04/2024 - Current

BeeSafe

- Applied computer vision to detect threats against honeybee colonies in apiculture, improving colony health indicators.
- Built a real-time camera system; achieved 84% detection accuracy after advanced dataset optimisation and cleaning.
- Developed a CNN-BiLSTM model for real-time bioacoustic threat detection, trained on a custom hive audio dataset; implemented sliding window buffering and post-processing smoothing to handle continuous audio streaming at inference.

15/05/2025 - 14/07/2026

Zen.AI

- Developed a comprehensive SaaS assessment & placement preparation portal tailored for institutions and recruitment partners.
- Built an AI-proctored test-taking platform and an AI-powered mock interview assistant leveraging resume/JD context.
- Delivered full end-to-end product: Email System, Calendar, Course Material, and Discussions modules.

08/01/2025 - 01/04/2025

WordLens

- Built an AI-powered reading aid using OCR and NLP to assist users with real-time reading comprehension and contextual word analysis.
- Designed word-level analysis pipeline with pronunciation support and contextual definitions.
- Patent published with India Patent Office – Application No. 202541122407 A

RESEARCH AND ACHIEVEMENTS

Government Research Grant – 10 Lakhs (≈ €11,000)

Awarded by the Indian Government for the Bee Threat Detection project, recognised at national level.

Patent Filed - Bee Threat Detection in Apiculture

A patent filed for the BeeSafe Project - **App. No. 202441071459 A**

Patent Filed - WordLens: AI-Assisted Reading System

A patent filed for the WordLens project - **App. No. 202561122407 A**

SKILLS

ML/AI

PyTorch · TensorFlow · Keras · scikit-learn · OpenCV · NLP · RAG

PROGRAMMING

Python · Flask · JavaScript · Node.js · React · Java

INFRA

AWS · GCP · Docker · Linux · Git

DATABASE

PostgreSQL · MongoDB · MySQL · Fiass

LANGUAGE SKILLS

MOTHER TONGUE(S): Telugu

OTHER LANGUAGE(S):

English

Listening C2

Reading C2

Writing C2

Spoken production C2

Spoken interaction C2

German

Listening A2

Reading A2

Writing A2

Spoken production A2

Spoken interaction A2